A minimax free-energy account of prioritized reactivation and systems consolidation in coupled predictive-coding networks

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Abstract

During sleep and quiet wakefulness, the brain spontaneously revisit neural activity patterns associated with previous experience, a phenomenon known as reactivation. Reactivation is considered to support systems consolidation, the process by which the retrieval of memory content becomes progressively more dependent on neocortical rather than hippocampal circuits. Evidence indicates that reactivation events are not indiscriminate but preferentially target specific memory traces.

Systems consolidation has motivated substantial modeling efforts, and existing models offer complementary accounts of spontaneous reactivation, the preferential replay of specific memory traces, and hippocampo-neocortical redistribution of memory representations. Here, we ask whether these phenomena, and their underlying neural implementation, can be jointly recovered from a **functionally motivated** normative objective.

We model the hippocampo-neocortical complex as two coupled associative memory networks implemented within the predictive coding (PC) framework. We then define our objective as minimizing the largest neocortical reconstruction deficit over the hippocampal memory manifold.

We show that, under suitable simplifying assumptions, the intrinsic dynamics of the two networks and their specific coupling implement gradient-based optimization of this objective. The architecture implements targeted memory reactivation and progressively redistributes memory representations through a closed-loop dialogue between the two systems. We also show that under these dynamics, quasi-exhaustive memory content transfer between the two systems is significantly faster than for random reactivation.

Moreover, this coupling circuitry aligns with biological data on the specific pathway and interactions modalities between the prefrontal cortex and hippocampal Cornu Ammonis 1.

Author summary

1 Introduction

An important fraction of the brain's metabolic budget is devoted to intrinsic activity, neuronal dynamics generated spontaneously by the brain in the absence of specific sensory input or behavioural demands [?, ?]. One of the best characterised events taking place during intrinsic activity is neural reactivation: the reinstantiation, during sleep or quiet rest, of activation patterns observed during waking experience (REF). For a substantial fraction of these events, activity in the hippocampus and neocortex is coordinated, giving rise to what is termed the hippocampo-cortical dialogue [?, ?].

This dialogue supports systems consolidation, the process by which memory retrieval becomes progressively less dependent on the hippocampus and more dependent on neocortical circuits [?, ?, ?, ?]. Systems consolidation is therefore often described as a form of memory transfer, whereby the neocortex progressively acquires memory content initially encoded in the hippocampus (REF).

Extensive literature indicates that the reactivation events underlying systems consolidation preferentially target certain memory traces over others. Traces that are recent, that correspond to novel experiences, or that **remain weakly encoded in cortex are prioritized for reactivation (REF)**. The precise neural mechanisms inducing this prioritization remain poorly understood (REF).

Reactivation and systems consolidation have motivated a broad range of computational models, from complementary learning systems [?] to autonomously reactivating attractor networks [?, ?, ?], generative replay [?, ?], and utility, context, or familiarity based replay scheduling [?, ?, ?, ?, ?]. These accounts differ in what drives reactivation, such as trace strength, recency, attractor competition, context, reward, or expected utility, and what information is transferred. Here, we focus on a single candidate mechanism, grounded in a specific communication motif between the hippocampus and neocortex.

The communication motif we consider has been most clearly characterized between the prefrontal cortex (PFC) and hippocampal area CA1, and combines two directional interactions: a bottom-up drive from CA1 to PFC and a top-down suppression from PFC to CA1. **During NREM sleep, a brief synchronized burst within a CA1 neuronal assembly is associated with increased firing in functionally coupled PFC populations (REF). Conversely, when PFC populations reactivate independently, without a coincident CA1 burst, CA1 neurons that normally participate in coordinated CA1–PFC events are preferentially suppressed (REF). Together, these findings motivate an asymmetric, representation-selective dialogue: hippocampal reactivation recruits selected prefrontal populations, whereas independent prefrontal reactivation suppresses the corresponding hippocampal representation. This motif may extend beyond PFC–CA1: entorhinal projections also recruit local inhibitory circuits in CA1, providing an potential additional pathway for similar bi-directional interation (REF).**(change place maybe)

Wang, Poe, and Zochowski gave this motif a functional interpretation: cortical memory representations regulate hippocampal reactivation according to their degree of consolidation [?]. In their model, as a cortical representation forms during reactivation, it recruits hippocampal inhibitory interneurons that selectively suppress the corresponding hippocampal trace. As consolidation proceeds, each trace withdraws its own reactivation, leaving replay to the content that still lacks cortical support.

We extend this account in three ways. First, we develop the functional role of feedback inhibition in organizing memory transfer. Second, we formulate the interacting hippocampal and neocortical memories within the predictive-coding framework. Third, and most fundamentally, we show that the resulting architecture and dynamics perform greedy optimization of a biologically plausible normative objective.

In the proposed model, neocortical feedback inhibition biases reactivation away from hippocampal content already supported cortically and toward content that remains poorly represented. This dynamic directs the consolidation process toward the largest representational gap between the two systems. We therefore show that, within the predictive-coding framework and under simplifying assumptions, the dynamics perform gradient descent on the largest cortical reconstruction deficit over the hippocampal memory manifold. The resulting architecture gives rise to prioritized reactivation, redistributing memory content as consolidation proceeds, and terminating reactivation once transfer is complete.

More broadly, this work contributes to the study of fast and reliable offline information transfer between artificial neural networks.

We deliberately adopt a simplified model in which the hippocampus and neocortex are each reduced to a single associative memory network of continuous activity units rather than spiking neurons; the resulting analytical tractability is what allows the mechanism to be derived rather than assumed. The analysis is performed on a linear version of the network, and the results are then generalized numerically for nonlinear activation functions.

2 Models

In this section, we introduce the network architecture, beginning with its local dynamics and their coupling. We then present the network training and memory consolidation dynamics.

The model consists of two coupled associative-memory predictive-coding networks (PCNs), the teacher and student networks, corresponding respectively to the hippocampus and neocortex. Taken independently, each network performs retrieval by carrying out inference on a given cue, minimizing its local free energy through changes in neuronal activity. Each network also learns by minimizing its local free energy, this time through changes in its synaptic weights (REF). The full architecture adds an asymmetric coupling between the teacher and student networks.

Memory storage occurs in two successive phases. The teacher first acquires externally imposed activity patterns; the coupled network then transfers this content to the student during intrinsic consolidation (Sec. 2.3).

2.1 A single predictive-coding associative memory

Tang et al. introduced the implicit covariance-learning predictive-coding network (implicit covPCN), which constitutes the basic building block of our architecture [**?**]. The implicit covPCN was developed as a more biologically plausible alternative to the explicit covPCN, whose learning rule requires non-local matrix operations and becomes numerically unstable on structured data [**?**, **?**, **?**, **?**]. Here we present the linear version of the network which is the basis of the analytical study presented in this work.

Both the teacher T and the student S, corresponding respectively to the hippocampus and neocortex, are implicit covPCNs with dense recurrent connectivity and no self-connections. Each performs associative memory through local inference and plasticity.

We use $\alpha \in \{S, T\}$ to index either network. For activity $x_\alpha \in \mathbb{R}^d$ and recurrent weights W_α, define the prediction-error operator and error

$$M_\alpha = I - W_\alpha, \qquad \varepsilon_\alpha = M_\alpha x_\alpha, \tag{1}$$

and the free energy the network minimizes,

$$F_\alpha^{\mathrm{rec}}(x_\alpha; W_\alpha) = \frac{\pi_\alpha}{2}\|\varepsilon_\alpha\|^2 = \frac{\pi_\alpha}{2}\|M_\alpha x_\alpha\|^2, \tag{2}$$

where $\pi_\alpha > 0$ is the precision of the prediction. We call F_α^{rec} the recurrent free energy because it accounts only for the network's internal prediction error generated by its recurrent connectivity. Network α lowers F_α^{rec} along two routes on two timescales: fast *inference* changes its activity, while slow *plasticity* changes its weights,

$$\tau_\alpha\, \dot{x}_\alpha = -\pi_\alpha\, M_\alpha^\top \varepsilon_\alpha, \qquad \dot{W}_\alpha = \eta\, \pi_\alpha\, \varepsilon_\alpha\, x_\alpha^\top, \tag{3}$$

with activity time constant τ_α and learning rate η.

The prediction may optionally pass through a pointwise nonlinearity f, giving $\varepsilon_\alpha = x_\alpha - W_\alpha f(x_\alpha)$. We take the linear case $f = \mathrm{id}$ throughout the analysis and return to the rectified case $f = \mathrm{ReLU}$ in Section 3.8. In addition, as introduced in the next section, the teacher activity is normalized such that $|x_T| = 1$ after each integration step.

Each network is trained according to the plasticity dynamics in Eq. (3), either during interleaved clamping or through intrinsic activity during system consolidation (2.3). In a fully trained network α, the stored content is the kernel of its error operator,

$$\mathcal{U}_\alpha = \ker M_\alpha = \{x \in \mathbb{R}^d : M_\alpha x = 0\}, \tag{4}$$

the network's memory manifold. Every stored pattern satisfies $M_\alpha x = 0$ and has zero recurrent free energy F_α^{rec} [**?**].

For the linear network, $\mathcal{U}_\alpha$ is a linear subspace. Any linear combination of stored memories is therefore a zero-energy fixed point. The network stores a flat memory manifold rather than a set of point attractors as in classical Hopfield networks (REF).

2.2 Coupling the teacher and the student

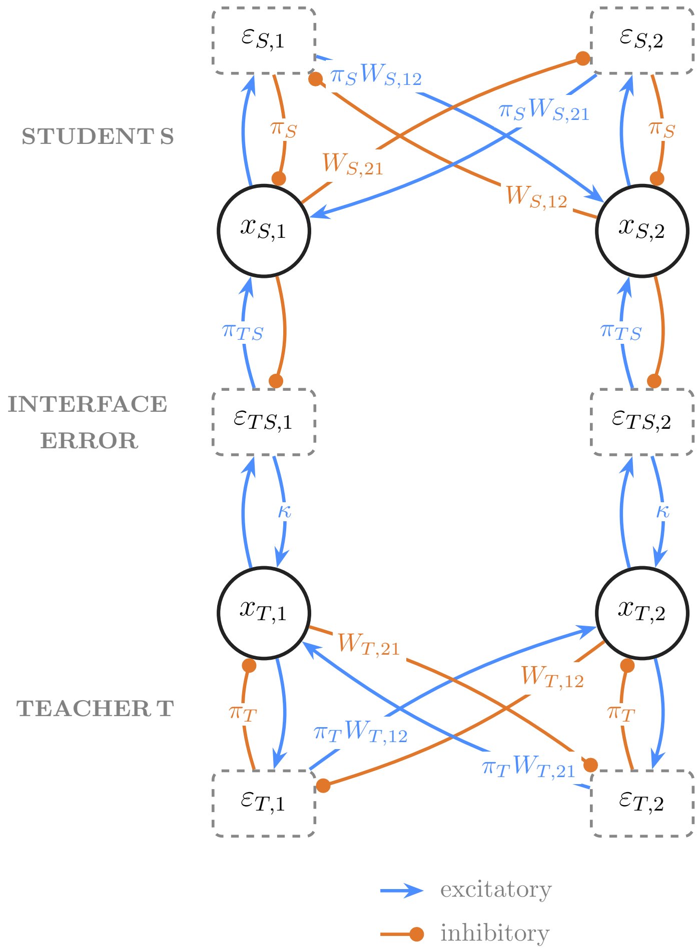

Fig 1. Model architecture (two units per population). Student S (top, cortical) above teacher T (bottom, hippocampal): value units x, recurrent errors $\varepsilon_S, \varepsilon_T$, and the one-to-one interface error $\varepsilon_{TS} = x_T - x_S$. Connections without labels have scalar value 1. During replay, the teacher-side interface gain κ makes the teacher ascend the student deficit rather than descending it as in classical PC networks.

Figure 1 shows the full architecture formed by coupling the student S and teacher T. The two networks interact through a one-to-one *interface error*,

$$\varepsilon_{TS} = x_T - x_S, \tag{5}$$

which measures the mismatch between the student state and the teacher state currently expressed. Without the interface influence, both the teacher T and the student S activities follow the inference dynamics in Eq. (3) and descends their own recurrent free energy F_T^{rec}.

For the teacher, the coupling adds a direct interface-error drive, giving

$$\tau_T \dot{x}_T = \underbrace{-\pi_T M_T^\top \varepsilon_T}_{-\nabla_{x_T} F_T^{\text{rec}}} + \kappa \varepsilon_{TS}, \qquad \|x_T\| = 1, \tag{6}$$

where κ is the teacher-side coupling gain. During intrinsic activity, $\kappa > 0$, so the interface-error population excites the teacher. In conventional predictive coding, minimizing the interface error with respect to x_T would instead contribute $-\pi_{TS}\varepsilon_{TS}$ to the teacher dynamics. The teacher-side pathway therefore reverses the conventional sign. After each integration step, teacher activity is normalized to unit norm to prevent the linear dynamics from producing unbounded activity. This operation idealizes fast population gain control: in biological circuits, divisive normalization mediated by recurrent and feedforward inhibition constrains the overall response magnitude while largely preserving the relative activity pattern that carries the represented content [**?**].

The student side retains the conventional predictive-coding dynamics. Student activity minimizes its recurrent free energy and receives an excitatory correction from the interface-error population,

$$\tau_S \dot{x}_S = -\pi_S M_S^\top \varepsilon_S + \pi_{TS}\varepsilon_{TS}, \tag{7}$$

the student activity is not normalized during the simulation of the whole model.

Both contributions to Eq. (7) are gradient-descent terms on the student free energy functional which can therefore be expressed as

$$F_S(x_S, x_T; W_S) = \underbrace{\tfrac{\pi_S}{2}\|M_S x_S\|^2}_{F_S^{\mathrm{rec}}} + \underbrace{\tfrac{\pi_{TS}}{2}\|x_T - x_S\|^2}_{F_{TS}}, \qquad \tau_S \dot{x}_S = -\nabla_{x_S} F_S, \tag{8}$$

where $\pi_{TS} > 0$ is the precision of the interface prediction error. The student weights follow the local plasticity rule

$$\dot{W}_S = \eta \pi_S \varepsilon_S x_S^\top = -\eta \nabla_{W_S} F_S. \tag{9}$$

Activity and plasticity evolve concurrently in the simulations.

The overall influence between S and T deserve carefull reading. Although $\kappa > 0$ denotes an excitatory connection from the interface-error population to the teacher, the student enters $\varepsilon_{TS} = x_T - x_S$ with a negative sign, inhibiting the interface error. The combined pathway $x_S \to \varepsilon_{TS} \to x_T$ is thus inhibitory: increasing student activity reduces the excitatory error drive reaching the teacher. This effective top-down inhibition is the model counterpart of prefrontal suppression of hippocampal activity. In a conventional predictive-coding network, the interface error would instead inhibit the lower population., producing top down excitation through disinhibition.

[small paragraph with place for citation about why it is bio-plausible to normalize teacher activity]

2.3 Initial storage and consolidation dynamics

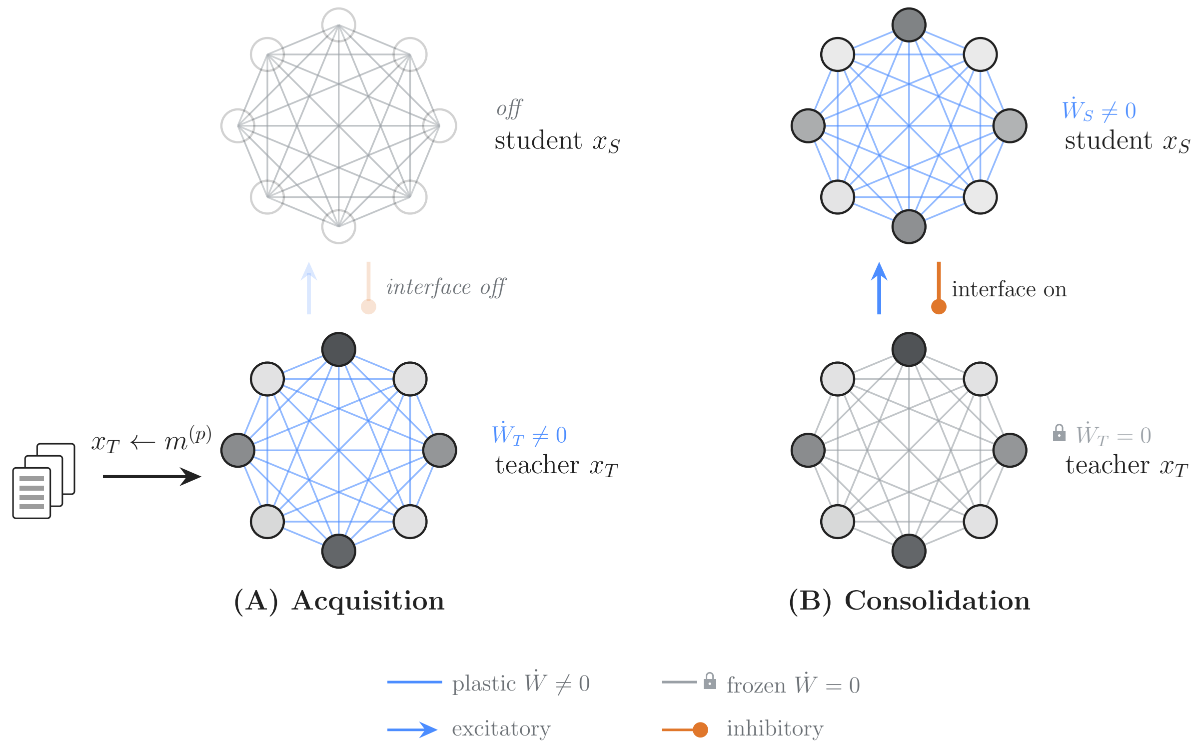

Fig 2. Initial storage and consolidation dynamics. Initial storage creates the teacher memory manifold from externally imposed patterns. Subsequent intrinsic dynamics use this manifold to generate student training states while preserving the teacher representation. The student is therefore trained by the coupled network dynamics rather than by direct clamping.

Memory is stored in the two networks by the same local recurrent plasticity rule (Eq. 3), but under two different regimes for generating neuronal activity. As shown in Fig. 2, the first phase establishes the teacher memory; the second transfers this content to the student without presenting the student with an external target.

During initial storage (Fig. 2(A)), teacher activity x_T is clamped to memory patterns $m^{(p)}$ in interleaved order while W_T follows Eq. 3. The student and the interface remain inactive. These updates reduce the teacher's recurrent prediction error on the presented patterns and establish its memory manifold $\mathcal{U}_T = \ker M_T$ (Methods). Interleaved training allows correlated memories to be stored while mitigating interference (REF). In contrast to our model, the hippocampus does not appear to require interleaved training to reliably store memory content during awake activity (REF). The orthogonalization of memory representations, particularly in the granule-cell layer of the dentate gyrus, is thought to support rapid learning of correlated memory traces (REF). This biological machinery is not part of our model and is therefore replaced by interleaved training.

During consolidation (Fig. 2(B)), the external clamp is removed and the interface is activated. The teacher and student activities follow Eqs. (6) and (7), while W_T remains fixed and only W_S follows the plasticity rule in Eq. (9). The student is never clamped: teacher-driven intrinsic activity supplies the states stored by its local recurrent plasticity and influence the student through the interface error.

3 Results

3.1 Reactivation and memory content transfer

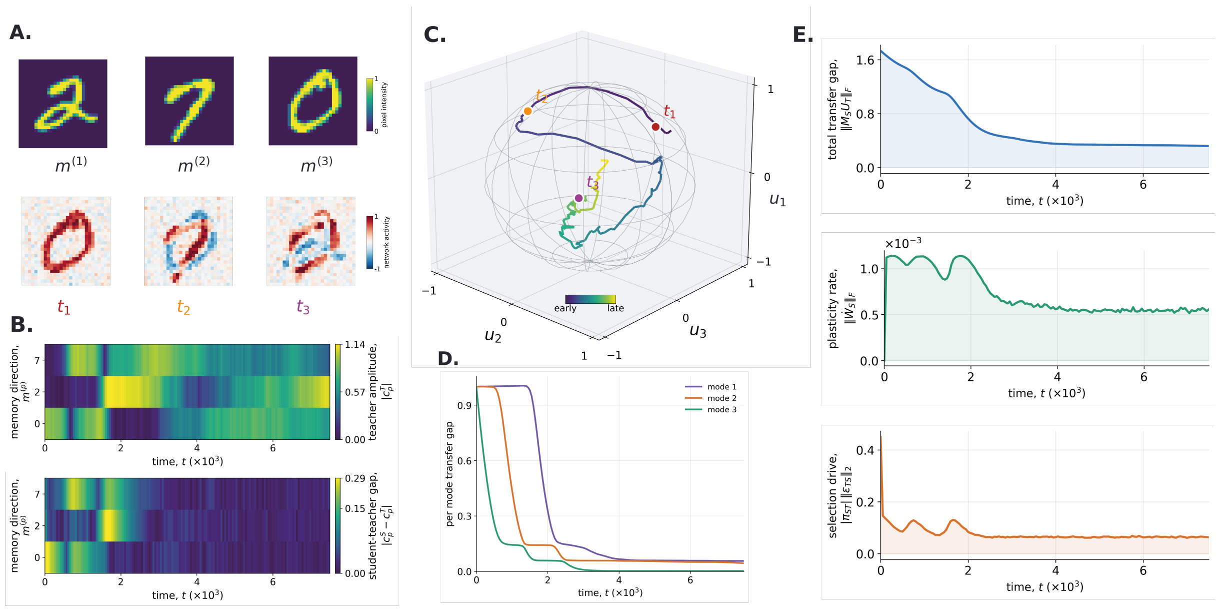

Fig 3. Spontaneous intrinsic activity progressively redistributes teacher-supported memory content. The frozen teacher was coupled to a student initialized with zero activity and zero recurrent weights. No external input was presented during the run; low-amplitude Gaussian noise seeded exploration of the teacher memory manifold. **A,** The three stored MNIST patterns (top; display order 2, 7, and 0) and teacher states at $t_1 = 250$, $t_2 = 1000$, and $t_3 = 7000$ (bottom). Although the stored patterns are positively signed, the linear network can express signed mixtures and antipodal states. **B,** (Top) Pattern amplitudes $|c_p^T(t)|$, which measure the contribution of each stored pattern $m^{(p)}$ to teacher activity $x_T(t)$. (Bottom) Student–teacher amplitude gaps $|c_p^S(t) - c_p^T(t)|$; a large gap indicates that the two networks express different amplitudes along that pattern coordinate. **C,** Teacher activity projected onto an orthonormal basis U_T of the three-dimensional teacher memory space. The wireframe marks the unit sphere imposed by teacher normalization, and the colored markers identify the snapshots in panel A. Basis axes are orthogonal memory-space directions, not individual digits. **D,** Sorted eigenvalues of $G_S(t) = U_T^\top M_S(t)^\top M_S(t) U_T$. High eigenvalues identify teacher-supported directions poorly represented by the student, whereas low values indicate accurate storage along those directions. **E,** (Top) Total transfer gap $\|M_S U_T\|_F$, which measures how much of the teacher memory space remains unsupported by the student. (Middle) Student plasticity rate $\|\dot{W}_S\|_F$, the magnitude of the current recurrent-weight update. (Bottom) Teacher-side selection drive $|\pi_{ST}| \|\varepsilon_{TS}\|_2$, the magnitude of the mismatch-dependent signal acting on the teacher.

We first examined whether intrinsic activity in our model could reactivate and redistribute stored memory content.

We stored three unit-normalized MNIST patterns in a linear 784-unit teacher network, froze the teacher's recurrent weights, and started the coupled dynamics with a plastic student whose recurrent weights were initialized to zero. During this intrinsic phase, neither network received external input. Low-amplitude Gaussian noise was applied to the teacher activity, allowing the dynamics to explore the teacher memory manifold. The results are shown in Fig. 3.

Fig. 3(A, bottom), (B, top), and (C) show that teacher activity is dominated by mixture states rather than by reactivation of individual patterns. This behavior follows from the linear activation function: the stored patterns span a flat memory manifold containing every signed linear combination of those patterns, including their antipodal states.

Fig. 3(D) and (E) provide direct evidence of memory-content redistribution during intrinsic activity. The successive reduction of the three eigenvalues indicates that the student acquires the flat teacher memory manifold in stages across independent directions, rather than through a uniform reduction of error. Transfer slows once most of the manifold is represented, but remains incomplete over the displayed interval. Longer simulations further reduce the residual gap, although at a much lower rate. At this late stage, the mismatch-driven selection signal is weak and the teacher increasingly relies on Gaussian-noise exploration to sample the remaining poorly represented directions. Despite this incomplete transfer, the isolated-student probes in Fig. 4 show that functional recall emerges early: clamped completion is uneven at the partial checkpoint but yields recognizable versions of all three memories by the late checkpoint.

Fig. 3(B, bottom) and (E, middle and bottom) show how the signals driving learning and reactivation attenuate during memory redistribution. The student plasticity rate decreases as the student–teacher mismatch contracts, revealing a self-limiting learning process: as the student acquires teacher-supported content, it removes the errors that generate further weight updates. In parallel, the teacher-side selection drive, generated by the mismatch-dependent top-down influence of the student on the teacher, also decreases. Three transient peaks accompany the successive reduction of the three error modes in Fig. 3(D), consistent with the sequential acquisition of the three independent directions spanning the rank-three teacher memory manifold. Both signals remain nonzero at the end of the displayed interval, so this run approaches but does not reach exact self-extinction.

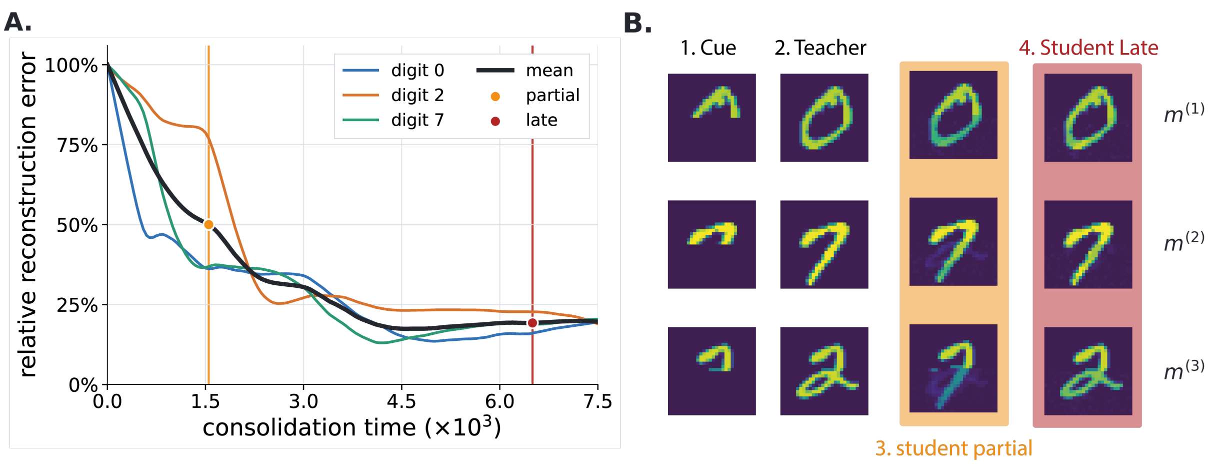

Fig 4. Transferred teacher memories become reconstructable by the isolated student. A, Relative recurrent reconstruction error for each stored MNIST pattern during consolidation. At each recorded time, the student weights were frozen, its state was set to the complete pattern $m^{(p)}$, and the error was evaluated as $r_p(t) = \|m^{(p)} - W_S(t)m^{(p)}\|_2/\|m^{(p)}\|_2$. Thus, 100% corresponds to the initially empty student with $W_S = 0$, whereas zero denotes exact recurrent prediction of that pattern. The black curve is the mean over the three digits; orange and red markers identify the partial ($t = 1550$) and late ($t = 6500$) checkpoints used in panel B. **B,** Clamped recall by isolated, frozen networks. From left to right, each row shows the partial cue, the teacher reconstruction, and student reconstructions at the partial and late checkpoints. The upper half of each digit was clamped to the stored pixels, the hidden half was initialized to zero, and the free units settled by minimizing the network self-energy $\frac{\pi}{2}\|(I - W)x\|_2^2$, with no teacher–student coupling or plasticity. Recall is uneven at the partial checkpoint, in agreement with the separated errors in panel A, but all three late outputs approach the teacher completions despite a remaining recurrent residual. All network reconstructions share a fixed viridis scale; negative activity is mapped to its lower endpoint.

3.2 The settled deficit identifies the least represented teacher-supported direction

$$S_S = M_S^\top M_S, \qquad N_S = \pi_S S_S(\pi_{TS} I + \pi_S S_S)^{-1}. \tag{10}$$

Proposition 1 (Settled deficit and spectral certificate). *For every $x_T \in \mathbb{S}_T$, the linear recipient has unique settled state*

$$x_S^* = (I - N_S)x_T, \qquad q(x_T; W_S) = \frac{\pi_{TS}}{2} x_T^\top N_S x_T. \tag{11}$$

If U_T is an orthonormal basis of $\mathcal{U}_T$ and $A_S = U_T^\top N_S U_T$, then

$$\mathcal{D}_{\max} = \frac{\pi_{TS}}{2}\lambda_{\max}(A_S). \tag{12}$$

[Explain that N_S measures unresolved content in the full state space and A_S restricts it to teacher-supported memories. The leading eigendirection is the largest current recipient deficit. Move scalar-mode and Rayleigh-Ritz proofs to S1 Appendix.]

3.3 Maximizing recipient deficit fixes the teacher-side interface drive

$$\nabla_{x_T} q = \nabla_{x_T} F_S\big|_{x_S = x_S^*} = +\pi_{TS}\varepsilon_{TS}^*. \tag{13}$$

$$\tau_T \dot{x}_T\big|_{\text{interface}} = \kappa\varepsilon_{TS}, \qquad \boxed{\kappa = \mu_T\pi_{TS} > 0.} \tag{14}$$

[Keep this short derivation in the main paper. Teacher dynamics expose the recipient's deficit, while recipient inference and plasticity reduce it. Do not reintroduce the removed saddle, zero-sum representation, or global free-energy claim.]

Theorem 1 (Local implementation of worst-case consolidation). *Under hard teacher-memory constraint, separated recipient-inference, teacher-selection and recipient-plasticity timescales, and a nonzero initial component in the leading discrepancy eigenspace, projected teacher dynamics ascend the settled recipient deficit and approach its maximizing eigenspace. At a simple leading eigenvalue, the subsequent local recipient update descends $\mathcal{D}_{\max}$ and gives the steepest feasible first-order decrease among sufficiently small updates of equal norm.*

Corollary 1 (Correlation-dependent local advantage). *Let two correlated unit memories span the teacher memory space and satisfy*

$$m_+ = cu + sv, \qquad m_- = cu - sv, \qquad c^2 = \frac{1+\chi}{2}, \qquad s^2 = \frac{1-\chi}{2}, \tag{15}$$

where u, v are orthonormal and $0 \le \chi < 1$. Suppose that the recipient already stores the common component but not the residual, $M_S u = 0$ and $M_S v \neq 0$, and consider unconstrained, exactly settled, infinitesimal plasticity. The worst-deficit selector then replays v. For replay of either named memory,

$$\dot{\mathcal{D}}_{\max}^{\text{named}} = \frac{1-\chi}{2}\dot{\mathcal{D}}_{\max}^{\text{priority}}, \tag{16}$$

where both derivatives are negative. Hence the ratio of positive instantaneous reduction rates is

$$\boxed{\frac{-\dot{\mathcal{D}}_{\max}^{\text{priority}}}{-\dot{\mathcal{D}}_{\max}^{\text{named}}} = \frac{2}{1-\chi}.} \tag{17}$$

The factor compares local derivatives at the current weights. It is not a ratio of complete transfer times and does not imply global convergence. The exact expression is not claimed after zero-diagonal weight projection.

3.4 Learning removes the signal that selected a memory

Corollary 2 (Completion and self-extinction)**.**

$$\mathcal{D}_{\max} = 0 \quad \iff \quad \mathcal{U}_T \subseteq \ker M_S. \tag{18}$$

At completion, $x_S^* = x_T$, $\varepsilon_{TS}^* = 0$, and $\varepsilon_S^* = 0$ for every valid teacher state. Both teacher selection and recipient plasticity vanish.

This corollary characterizes the completed state and the disappearance of its local learning signals. It does not establish that the weight dynamics reach that state from arbitrary initial conditions.

[Connect this to the biological claim: learning removes the discrepancy that generated priority, without a separate decay variable.]

3.5 Recipient deficits organize priority and recipient-relative familiarity

[Show target switching: the teacher first occupies the largest deficit, then shifts after that direction is learned. Compare naive, partially familiar, and fully familiar recipients with the teacher fixed. State that priority falls with recipient competence conditional on source support; source reliability remains a distinct variable.]

[figure placeholder]
Fig 3. Priority, target switching, and familiarity. Panel A: teacher-restricted novelty eigenvalues. Panel B: direction-by-bout teacher occupancy. Panel C: plasticity and reactivation probability. Panel D: naive, partially familiar, and fully familiar recipients.

3.6 Differential waking learning prioritizes successfully encoded weak memories

[During waking, use fast hippocampal and slow cortical plasticity to learn one orthogonal memory extensively and a second only until its hippocampal reconstruction error is low while its cortical error remains high. With order counterbalanced, test whether subsequent offline dynamics preferentially reactivate the second, source-supported but cortically weak memory.]

[figure placeholder]
Fig 4. Waking learning strength and subsequent reactivation. Panel A: exposure protocol. Panel B: hippocampal and cortical reconstruction errors before sleep. Panel C: memory-specific reactivation occupancy during subsequent offline consolidation.

3.7 Autonomous prioritization produces faster transfer than random reactivation in simulations

[Primary empirical efficiency test. Under the common discrete protocol, determine whether autonomous selection approaches the greedy oracle while reducing the worst-case deficit more efficiently than random teacher-memory selection. Sweep memory correlation and compare the residual-phase initial rate with Eq. (17) in the unconstrained condition. Treat events to threshold and final transfer quality as simulation results, not consequences of a global timing theorem.]

[figure placeholder]
Fig 5. Prioritized versus oracle versus random transfer. Panel A: $\mathcal{D}_{\max}$ versus identical plasticity events. Panel B: residual-phase first-order reduction versus memory correlation, with the unconstrained prediction $2/(1-\chi)$. Panel C: $\mathcal{D}_{\max}$ versus cumulative recipient-weight change. Panel D: events to fixed thresholds and autonomous-to-oracle selection gap. Panel E: final transfer quality with matched-seed uncertainty.

3.8 Robustness and rectification

[Main text: summarize reliability across tested memory ranks, dimensions, correlations, partial familiarity, timescale separations, noise levels, and teacher-manifold leakage. Supporting Information: full sweeps. Use "reliable across tested regimes," not a global convergence claim.]

[For ReLU without bias and non-negative stored patterns, show early trajectories near individual memories, later weakening as they are learned, time-resolved nearest-memory distance, cone confinement, and honest final-transfer quality.]

[figure placeholder]
Fig 6. Rectified dynamics and individual-memory reactivation. Panel A: linear signed-mixture geometry versus non-negative ReLU cone. Panel B: early and late trajectories. Panel C: time-resolved nearest-memory distance. Panel D: cone confinement. Panel E: transfer quality and self-extinction relative to the linear network.

4 Discussion

However, CA1–PFC representational coordination was weaker and less faithful to the waking experience during sleep than during awake ripples. The authors interpreted sleep reactivation as more integrative and less like precise reinstatement. CLEAR LIMITATION DUE TO THE FACT THAT WE USES SIMPLE ASSOCIATIVE MEMORY NETWORKS

4.1 Summary

[One paragraph: recipient deficit selects teacher content; selected content trains the recipient; learning removes the selecting signal. Prioritization, reduced reactivation of consolidated content, and self-extinction follow from the same objective.]

4.2 Relationship to other models

[Compare utility-based replay, context-driven replay, autonomous attractor consolidation, generative consolidation, recall-gated consolidation, and Go-CLS by their priority variable, autonomous selection mechanism, and what is transferred. The narrow novelty claim is a closed loop from recipient-deficit objective to source selection and recurrent recipient storage.]

4.3 Awake prefrontal memory control may be reused during sleep

[Lead with awake retrieval suppression and retrieval-induced forgetting. During wake, PFC control can suppress unwanted or competing hippocampal retrieval. Propose that related effective circuitry is reused offline: behavioral goals select during wake, whereas recipient competence selects during consolidation. Link to NREM PFC-associated suppression of CA1 activity and reactivation as evidence for a suppressive motif, not proof that PFC computes $\mathcal{D}_{\max}$.]

4.4 Experimental predictions, limitations, and future directions

[Predictions: cortical similarity suppresses matching hippocampal reactivation after controls for source strength, recency, salience, and reward; disrupting top-down suppression increases redundant reactivation and reduces efficiency; priority returns after cortical degradation. Limitations: linear subspaces; restricted ReLU extension; no ordered replay; idealized constraints and timescales; degeneracy; transformed codes; omitted source reliability, context, reward, emotion, salience, and neuromodulation. State explicitly that the analytic efficiency claim is local, that the exact $2/(1-\chi)$ factor assumes unconstrained weights and an already stored common component, and that faster complete transfer is supported numerically rather than by a global convergence or hitting-time theorem. Continual learning is a separate downstream project.]

5 Conclusion

[Two or three sentences. Close on the biological and mathematical message: cortical dependence can organize hippocampal reactivation; effective cortical suppression of the teacher follows from maximizing recipient deficit; and the discrepancy-driven prioritization signal vanishes at complete transfer.]

6 Materials and methods

6.1 Memory construction and network initialization

[Pattern generation, covariance-memory construction, dimensions, rank, correlation, recipient familiarity, seeds, and no-autapse constraint.]

6.2 State dynamics and diagnostics

[Teacher and recipient equations, activation, integration scheme, timestep, timescales, coupling, noise, normalization, settling, stopping, manifold leakage, and terminal occupancy.]

6.3 Prioritized, oracle, and random protocols

[Executable pseudocode or an algorithm box. Matching initializations, candidate states, one plasticity event per bout, thresholds, seed count, and uncertainty summaries.]

6.4 Outcome measures

[Define N_S, A_S, $\mathcal{D}_{\max}$, errors, occupancy, cumulative weight change, threshold events, oracle gap, nearest-memory distance, and cone confinement.]

Supporting information

- **S1 Appendix.** Full proofs of Proposition 1, Theorem 1, Corollary 1, and Corollary 2; projected power iteration; envelope steps; antipodal maximizers; feasible projected descent; and the local correlated-memory rate comparison.
- **S1 Fig.** Ascent, zero-drive, and descent teacher-side coupling ablation.
- **S2 Fig.** Rank, dimension, correlation, and recipient-familiarity sweeps.
- **S3 Fig.** Timescale separation, noise, seed dependence, and teacher-manifold leakage.
- **S4 Fig.** Oracle implementation checks and random-baseline controls.
- **S5 Fig.** Rectified-network robustness and time-resolved individual-memory selectivity.